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RESEARCH ARTICLE

Robust Driver State Assessment on an Embedded Platform Using a Fuzzy Belief Rule-Based Fusion of Eye Aspect Ratio and Voice Commands

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ABSTRACT This research presents an Adaptive Driver Monitoring and Assistance System (ADMAS) built on a multimodal AI architecture to prevent accidents caused by critical driver states such as drowsiness, fatigue, and distraction. At its core, a Fuzzy Belief Rule-Based (FBRB) inference engine integrates multimodal data from camera and microphone sensors to ensure robust decision-making. The visual stream computes an innovative EAR4 feature derived from facial landmarks, achieving 89.4% recall in machine learning classification tasks. To enhance robustness, the system establishes a personalized adaptive threshold for each driver during an initialization phase by analyzing data from over 300 frames. Concurrently, a speech recognition module enables proactive, system-initiated assistance via a voice interface, allowing the system to deliver vocal prompts and manage functions based on conversational input, thereby minimizing driver distraction. The FBRB engine fuses the EAR-4 feature, head pose estimations, and voice intent to assess the driver's state, subsequently triggering timely drowsiness alerts and autonomous system adjustments. The system, deployed on an NVIDIA Jetson Orin Nano for low-latency on-device processing, achieved reliable detection at 25 frames per second (FPS) under dynamic driving conditions, confirming its viability for real-world deployment.

INDEX TERMS Artificial intelligence, belief rule based system, eye aspect ratio, fatigue detection, fuzzy logic, machine learning, MediaPipe, whisper.

I. INTRODUCTION

Driver fatigue and distraction are widely recognized as critical factors in vehicular accidents, which frequently result in severe injuries and fatalities. Indeed, extensive research consistently identifies these states as primary causes of traffic-related mortality and morbidity [1], [2], [3]. For instance, the National Highway Traffic Safety Administration (NHTSA) estimates that drowsy driving accounts for approximately 100,000 crashes, over 1,500 fatalities, and 71,000 injuries annually in the United States alone [4]. A compound issue is that drivers often exhibit poor self-assessment of their fatigue levels, underestimating the degree of their impairment. This lack of self-awareness poses a significant

risk not only to the compromised driver but also to other road users. Consequently, the development of systems capable of real-time driver state monitoring is paramount to enhancing road safety and preventing such incidents [5], [6], [7]. Driven by progress in sensing and computational technologies, a variety of methods for the automated detection of driver fatigue have been proposed in the literature. These approaches can be broadly categorized as follows.

A. PHYSIOLOGICAL SIGNAL BASED APPROACH

These techniques rely on the measurement of physiological parameters such as heart rate, electroencephalography (EEG), and ocular movements to assess the driver's alertness level [8], [9]. The primary advantage of this approach lies in its high accuracy, which is achieved through direct

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physiological measurement. However, a significant limitation is the requirement for specialized equipment, which can be invasive to the driver in real-world scenarios.

B. VEHICLE BEHAVIOR BASED APPROACH

By analyzing driving behavior such as steering wheel activity, lane departure patterns, and pedal usage researchers can infer the driver's attention level. These methods are non-intrusive and do not require direct interaction with the driver. However, their effectiveness can be influenced by external factors such as road conditions, traffic flow, and weather [10].

C. VISION BASED APPROACH

Computer vision techniques have been applied to monitor visual cues, including eye closure frequency, head pose, and facial expressions [5], [6]. The non-contact nature of these systems makes them particularly attractive for real world deployment. Nevertheless, their performance may be hindered by ambient lighting, camera placement, or driver occlusion [11].

D. MULTIMODAL FUSION APPROACH

To improve robustness and reduce the limitations of single-modality systems, recent studies have explored multimodal fusion strategies [13]. By integrating visual features with physiological signals or combining vehicle dynamics with environmental context, these systems aim to enhance detection reliability and adaptability across diverse driving scenarios [5], [6], [12], and [14].

While the aforementioned methods demonstrate strong performance against technical metrics, they give little consideration to the User Experience, resulting in a deficit of effective communication between the system and the user. This study introduces a non-invasive driver monitoring framework that integrates visual and auditory sensing, based on an adaptation of the Fuzzy Belief Rule-Based Approach [3]. The feasibility of robust auditory sensing in noisy in-vehicle environments is supported by recent advancements in large-scale models like Whisper [23]. Specifically, the system integrates a camera and microphone for real-time data acquisition and analysis, implemented on an edge computing platform to minimize latency and dependency on external infrastructure. This approach, referred to as Adaptive Driver Monitoring and Assistance System (ADMAS), is designed to continuously evaluate driver fatigue and distraction through multimodal cues. However, achieving reliable and real-time inference in real-world environments remains a non-trivial challenge. It necessitates addressing key issues such as processing efficiency, computational constraints of embedded hardware, and long-term system stability. Addressing every type of uncertainty in driver fatigue assessment continues to be a difficult area of research. While traditional IF-THEN rules struggle to grasp the complex causal connections and uncertainties between different factors, Belief Rule-Based Approach (BRBA) effectively manage these

complexities [12], [15], [16]. Within BRBA [17], rule bases are modeled using the IF-THEN structure, and the mediapipe algorithm is employed for inference purposes. The remainder of this paper is organized as follows. Section II reviews the related work. Section III describes the proposed methodology, which integrates a belief rule-based approach with image and voice recognition algorithms. Section IV presents the experimental results from the implementation of the overall architecture on an edge platform. Section V discusses the results and their implications. Finally, Section VI concludes the paper.

II. RELATED WORK

Driver Monitoring Systems (DMS) based on computer vision have become a critical area of research due to their significant role in enhancing automotive safety. To improve system robustness and accuracy, hybrid models that combine head pose and eye state analysis have been proposed to overcome the inherent limitations of single-modal approaches. This is because methods focusing solely on head pose often require multiple cameras or complex 3D face models for accurate estimation, while techniques that analyze only eye-related features, such as gaze direction, tend to be effective only when the driver's face is oriented toward the camera. Numerous advanced hybrid models have been developed. For instance, in [18] presented a real-time monitoring system that employs a hybrid neural architecture to extract spatiotemporal features from monocular camera sequences to detect driver fatigue and distraction, achieving a processing speed of 15 frames per second. Similarly, F. Vicente et al. introduced a real-time, vision-based system that requires no manual calibration to detect Eyes Off the Road glances. Their hybrid model demonstrated stable performance across various lighting conditions and drivers, reaching a maximum processing speed of 25 FPS.

To further enhance detection reliability, some research has integrated vehicle dynamics data. In [19] proposed a deep learning framework that combines facial features, processed using the VGG-Face model, with vehicular data such as speed and steering angle extracted from the Controller Area Network (CAN) bus. Following a similar approach, the system simultaneously processes [20] CNN-analyzed facial cues and steering behavior to evaluate driver fatigue. Their system triggers an immediate alert upon detecting drowsiness and demonstrated a detection accuracy of 92.5% at a processing speed of 23 FPS. Beyond vision and vehicle data, other research has explored alternative sensing modalities. Several studies have investigated the use of physiological signals, such as heart rate variability, in conjunction with fuzzy logic-based driver monitoring systems (FDMS) to detect abnormal driving behaviors. The integration of fuzzy inference with machine learning algorithms has been shown to improve classification performance and decision-making reliability, offering a more nuanced understanding of driver states. Notable examples of such approaches can be found in [21] and [22]. The development of computer vision and voice-based algorithms is an essential component of edge

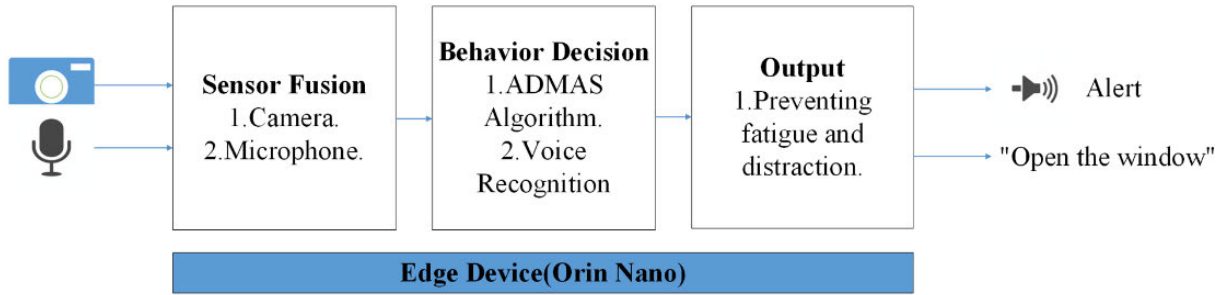


FIGURE 1. Overall architecture of the proposed adaptive driver monitoring and assistance system.

computing workflows for real-time DMS deployment. One of the most critical challenges in this process is ensuring that the system undergoes comprehensive testing and validation across diverse driving conditions. To achieve a fully integrated Advanced Driver Assistance System (ADAS) that combines visual and auditory modalities, several design and engineering considerations must be addressed throughout the development pipeline.

This study bridges academic research and industrial implementation in the domain of automotive safety. The proposed system integrates theoretical foundations, algorithmic development, practical engineering methods, and real-time edge deployment strategies, all within a unified and cohesive design framework. The final implementation has been successfully developed and validated through extensive testing, demonstrating its feasibility for real world application.

III. METHODOLOGY

Safe driving requires continuous attention and vigilance. Among the most common forms of inappropriate driving are distracted driving and fatigued driving, both of which pose significant threats to road safety and are major contributing factors to traffic accidents. However, these risky behaviors are largely preventable with the aid of intelligent monitoring systems. To address this challenge, we propose the design of an Adaptive Driver Monitoring and Assistance System (ADMAS), which integrates artificial intelligence models with a Fuzzy Belief Rule-Based (FBRB) inference mechanism [3], [12]. The system leverages data from both camera and microphone sensors to implement a real-time, multimodal fusion algorithm capable of adapting to dynamic driving conditions.

By detecting early signs of fatigue and distraction, the system aims to provide timely warnings and effectively reduce the likelihood of accidents. The overall system architecture of ADMAS, including sensor fusion, behavior decision, and output mechanisms, is illustrated in Fig. 1. This diagram highlights the multimodal integration process and its deployment on an edge device. In the following sections, we present the detailed structure of the proposed architecture, including the theoretical foundations and operational principles of each component model.

A. BELIEF RULE-BASED APPROACH

The Belief Rule-Based Approach (BRBA) utilizes a knowledge base constructed from belief rules and is characterized by four core components: input transformation, rule activation weight evaluation, belief degree updating, and rule aggregation [3], [17]. Unlike conventional data processing methods that presume well-structured problems and complete information, real-world decision-making must contend with inherent uncertainty. Such uncertainty arises from ambiguous data, incomplete evidence, and observational inaccuracies. The detection of driver drowsiness is a prime example of such a problem, as it manifests through subtle and often ambiguous changes in facial features like the eyes and mouth. Consequently, visual-based detection methods require robust algorithms capable of interpreting these uncertain characteristics.

In contrast to traditional IF-THEN rules, which struggle to capture the complex uncertainties among various factors, the Belief Rule-Based Approach (BRBA) is specifically designed to address this challenge. A belief rule's structure is composed of two primary components: an antecedent and a consequent [12]. The antecedent section contains one or more attributes and explicitly models uncertainty through parameters such as rule weights, attribute weights, and degrees of belief. The consequence, in turn, corresponds to a single conclusion attribute. A single belief rule can be formally expressed by the following equation:

$$R_k = \begin{cases} IF (P_1 \text{ is } A_{1k}^k) \cap (P_2 \text{ is } A_{2k}^k) \cap \dots \cap (P_N \text{ is } A_{Nk}^k) \\ Then \{(C_1, \beta_{1k}), (C_2, \beta_{2k}), \dots (C_N, \beta_{Nk})\} \end{cases} \quad (1)$$

The k_{th} rule, associated with antecedent attributes $P_1, P_2, \dots$ is defined by a belief degree $\beta_{ik} (\beta_{ik} \geq 0, \sum_{j=1}^N (\beta_{jk}) \leq 1)$, a rule weight q_k , and attribute weights $(\delta_{k1}, \delta_{k2}, \dots, \delta_{kNK})$. Due to ignorance or incomplete information, the rule may be incomplete, in which case the sum will be less than 1. The calculation formula is expressed as follows:

$$L = \prod_{i=1}^T (j_i), \quad (2)$$

This phase involves calculating both the combined matching degree and the weights of the rules. Despite a rule having

either single or multiple antecedent attributes as shown in (1), the initial task is to determine the combined matching degree necessary for computing the rule's activation weight. This calculation can be performed using a multiplicative function, outlined in (3), that enables integration among the antecedents of the rule.

$$\alpha_k = \prod_{i=1}^{T_k} (\alpha_i^k)^{\bar{\delta}_{ki}}, \bar{\delta}_{ki} = \frac{\delta_{ki}}{\max_{i=1, \dots, T_k} (\delta_{ki})}, \quad (3)$$

Ensuring $0 \leq \delta_{k1} \leq 1$, δ_{ki} ($i = 1, \dots, T_k$) is the proportionate weight of the i th antecedent attribute in the k th rule, calculated by dividing its weight by the greatest weight among all the antecedents.

In the Belief Rule Base Approach (BRBA), the activation weight (ω_k) signifies the importance of each rule, computed using the predefined matching degree according to equation (4):

$$\omega_k = \frac{\theta_k \alpha_k}{\sum_{i=1}^k \theta_i \alpha_i}, \quad (4)$$

where θ_k is the relative weight of the k th rule [17].

When the activation weight ω_k equals 0, the rule does not affect the belief rule base; conversely, $\omega_k = 1$ demonstrates that the rule has a significant influence. For a rule, if the sum of the degree of belief is less than zero, it renders the rule incomplete. In such cases, the belief degree needs to be updated by using:

$$\beta_{ik} = \bar{\beta}_{ik} \frac{\sum_{i=1}^{T_k} (\tau(t, k) \sum_{j=0}^{j_i} \alpha_{ij})}{\sum_{i=1}^{T_k} (\tau(t, k))}. \quad (5)$$

where, $\tau(t, k)$ is the membership degree of the k -th rule's output set, is the k -th rule's output value, and T is the total number of rules.

B. 3D FACIAL FEATURE EXTRACTION APPROACH

The foundation of the proposed system is high-precision 3D facial landmark detection. To this end, we employ Google's MediaPipe framework, which has been validated for its efficiency and accuracy in numerous real-time machine learning applications. Specifically, we utilize its Face Mesh module, which integrates the lightweight BlazeFace detector, to extract a dense mesh of 468 3D facial landmarks from each video frame. Compared to traditional 2D landmarks, these 3D points provide rich depth and spatial information, establishing a more robust data foundation for subsequent pose estimation and state analysis.

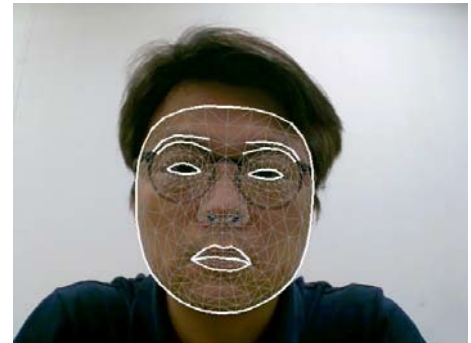
Once the 3D facial landmarks are obtained, the system proceeds to estimate the head pose to determine the driver's direction of attention. For this step, we adopt the methodology proposed by Ming et al. [6], selecting a specific subset of six key points to construct a head coordinate system. By establishing the correspondence between these 3D landmarks and their 2D projections on the image plane, the Perspective-n-Point (PnP) algorithm is then applied to solve for the head's rotation and translation matrices relative to the camera [24].

The final output is the head's 3D orientation expressed in Euler angles, providing a quantitative basis for distraction detection. The state of the driver's eyes is one of the most direct indicators of fatigue. This module utilizes the previously detected eyelid landmarks to calculate the Eye Aspect Ratio (EAR) in real-time. EAR is a widely used metric that quantifies the level of eye openness by computing the ratio of the vertical and horizontal distances of the eyelid landmarks. When the EAR value drops and remains below a predefined threshold for a sustained period, the system identifies this as a potential state of drowsiness. As illustrated in Fig. 2, MediaPipe's Face Mesh module is capable of outputting up to 468 three-dimensional facial landmarks, which precisely capture the geometric structure and critical regions of the face, such as the contours of the eyes, eyebrows, nose, and mouth. We adopt the specific six key landmarks are selected the nose tip, chin, left eye corner, right eye corner, left mouth corner, and right mouth corner, denoted as A_1 through A_6 to construct a head coordinate system [6]. By mapping these six known 3D model points to their corresponding 2D projections on the captured image, we solve for the head's rotation and translation vectors using the PnP algorithm, as shown in (6):

$$\begin{aligned} A_1 &= (0, 0, 0) \\ A_2 &= (0, -330, -65) \\ A_3 &= (-225, 170, -135) \\ A_4 &= (225, 170, -135) \\ A_5 &= (-150, -150, -125) \\ A_6 &= (150, -150, -125) \end{aligned} \quad (6)$$



(a)



(b)

FIGURE 2. Mediapipe face mesh output of 468 3D facial landmarks. (a) without glasses (b) with glasses.

This estimation produces three orientation angles, namely pitch (vertical rotation), yaw (horizontal rotation), and roll (lateral tilt), which are critical for evaluating the driver's attention and gaze direction. By establishing the geometric correspondence between the 3D facial reference points and their 2D projections on the image plane, the head pose can be estimated using the Perspective-n-Point algorithm. The resulting rotation and translation vectors of the head relative to the camera are obtained as expressed in Equation (7).

$$\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad (7)$$

where (u, v) are the 2D image coordinates, (X, Y, Z) are the corresponding 3D model points, (f_x, f_y) represent the focal lengths, and (c_x, c_y) denote the optical center. The camera intrinsic matrix is defined as:

$$\text{Camera matrix} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \quad (8)$$

where f_x, f_y, c_x, c_y are calibrated using the method proposed by Zhang [25], which estimates the camera's internal parameters and lens distortion coefficients from a series of known reference patterns. Our proposed system utilizes MediaPipe FaceMesh to detect facial landmarks and applies the Perspective-n-Point (PnP) algorithm to estimate head pose. The derived rotation vector is transformed into Euler angles pitch, yaw, and roll, providing a quantitative representation of the driver's orientation.

In addition, to monitor signs of driver fatigue, the system utilizes the Eye Aspect Ratio (EAR). The EAR is a geometric metric derived from the relative distance between specific upper and lower eyelid landmarks. As detailed in the process shown in Fig. 3, high-precision eye region landmarks are extracted using MediaPipe's Face Mesh and Iris modules. The resulting EAR value provides a continuous indication of the eye's openness, making it highly sensitive to changes caused by blinking or drowsiness. As illustrated in Fig. 4, the Face Mesh module estimates six key eyelid landmarks for each eye, starting from the outer corner P_1 and proceeding clockwise along the eyelid contour to P_6 . Based on these points, the EAR is computed using the following formula [5]:

$$\text{EAR} = \frac{\|P_2 - P_6\| + \|P_3 - P_5\|}{2 \|P_1 - P_4\|} \quad (9)$$

To reduce computational complexity and focus specifically on upper eyelid motion associated with fatigue-related eye behavior, this study adopts a modified four-point Eye Aspect Ratio (EAR-4) formulation. Unlike the traditional six-point EAR that considers both upper and lower eyelids, the proposed method estimates vertical openness solely based on the upper eyelid's deformation.

As shown in Equation (10)-(14), the horizontal distance H is calculated between the outer and inner corners of the

eye, defined as $P_1(x_1, y_1)$ and $P_4(x_4, y_4)$, respectively. The vertical midline y_h is the average y-coordinate of these two points. Two vertical distances V_1 and V_2 are measured from the upper eyelid landmarks P_2 and P_3 to y_h . The modified EAR-4 is then computed as:

$$H = \sqrt{(x_4 - x_1)^2 - (y_4 - y_1)^2}, \quad (10)$$

$$y_h = \frac{y_1 + y_4}{2}, \quad (11)$$

$$V_1 = |y_2 - y_h|, \quad (12)$$

$$V_2 = |y_3 - y_h|, \quad (13)$$

$$\text{EAR-4} = \frac{V_1 + V_2}{2 \cdot H}. \quad (14)$$

This method, despite relying on only a few key landmarks, is capable of effectively capturing the state of eye openness, demonstrating both accuracy and robustness in practical applications, as illustrated in Fig. 5.

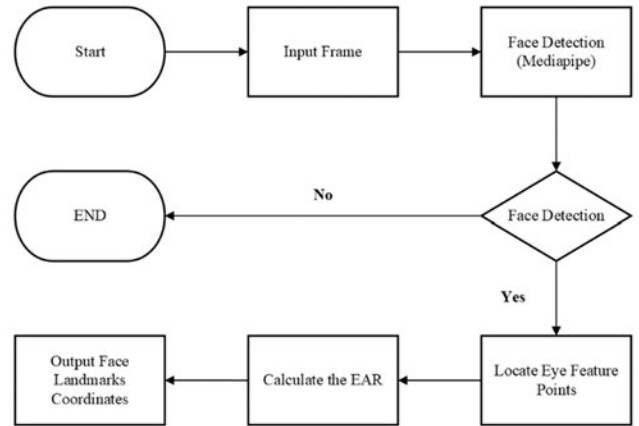
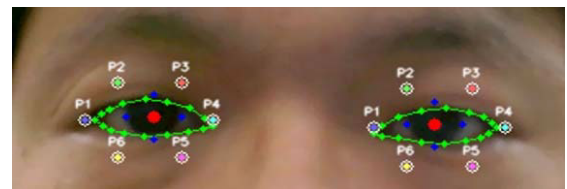
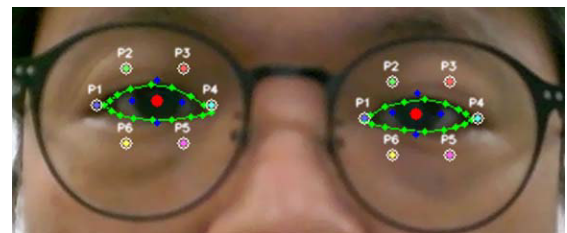


FIGURE 3. Flowchart of the eye aspect ratio (ear) calculation process using mediapipe.



(a)



(b)

FIGURE 4. Eye landmark Detection Using Mediapipe Under Different Conditions. (A) Without Glasses. (B) With Glasses.

C. AN APPROACH FOR VOICE COMMAND DETECTION

This study employs the Whisper-tiny speech recognition model developed by OpenAI for voice activity detection (VAD) as well as wake word and command detection tasks [23]. As a lightweight variant of the original Whisper architecture, it has been widely adopted in various speech recognition applications owing to its strong generalization across diverse languages and its robustness to background noise [23], [26]. The primary objective of this work is to further evaluate Whisper-tiny's performance specifically in VAD and wake word/command detection scenarios on resource-constrained edge devices. Regarding wake word and command detection, the model is fine-tuned on domain-specific datasets to improve its sensitivity to predefined voice commands or short phrases (e.g., “Hi Car”, “Open the Window”). By applying keyword spotting techniques to the text transcriptions produced by Whisper-tiny [23], the system can promptly identify wake words or commands and trigger the corresponding voice interaction modules.

The overall processing pipeline of the proposed Whisper-tiny voice interaction framework is illustrated in Fig. 6. First, the input audio undergoes signal enhancement using HTdemucs-based denoising to improve quality [27], [28], and [34]. Aiming for acoustic robustness, the framework utilizes HTdemucs to suppress non-stationary cabin interferences, such as road friction, HVAC airflow, and background music. The enhanced audio is then evaluated by the Whisper-tiny VAD module. If speech activity is detected, the signal proceeds with the Whisper-tiny model for transcription. Following transcription, the textual output is analyzed to detect the presence of a wake word or command. Once identified, the system activates the corresponding command function module; otherwise, the process terminates and awaits new audio input. This framework is engineered for two primary objectives: robustness and low latency. To achieve robustness, the initial Demucs noise suppression stage is essential, boosting the success rate of speech recognition in challenging, real-world acoustic conditions. Meanwhile, low latency is accomplished through a VAD module, which functions as a smart computational gatekeeper. By discarding silent and non-speech audio clips upfront, it prevents the computationally expensive Whisper-tiny model from wasting resources on irrelevant input, leading to a marked reduction in system delay. Consequently, this hierarchical and optimized workflow results in a system that is not only remarkably responsive but also holds immense promise for practical deployment on edge devices with limited computational power.

IV. EXPERIMENT RESULTS

This study presents an experimental multimodal driver state recognition framework that integrates both visual and auditory signals to detect signs of fatigue and inattentiveness, with a focus on practical deployment under edge computing constraints. The system architecture is composed of four key

components. First, the performance of two eye aspect ratio feature sets EAR and the simplified EAR4 is evaluated in the task of classifying open versus closed eye states. These features are tested across datasets of varying sizes using Random Forest and XGBoost classifiers to compare classification accuracy and computational efficiency [7], [29]. Second, head pose estimation (HPE) is incorporated into the visual module. By computing yaw and pitch angles from facial landmarks, the system is able to identify behavioral cues such as head tilting or downward posture, which may indicate attentional lapses that are not fully captured through eye features alone. Third, the integration of the Whisper-tiny automatic speech recognition module is introduced. By synchronizing the semantic content of spoken driver commands with real-time eye status, the system enables cross-modal comparison to enhance understanding of driver intent and detect inconsistencies between verbal claims and physiological signals. Finally, a fuzzy rule-based inference mechanism is implemented to integrate EAR values and head pose

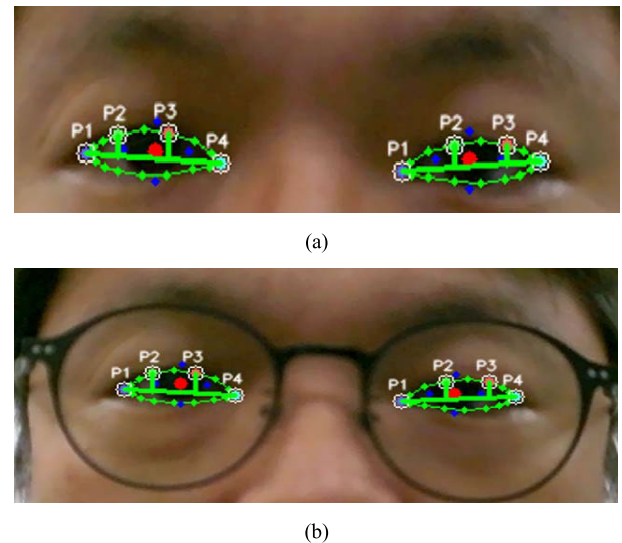


FIGURE 5. Illustration of 4-point eye aspect ratio (EAR-4) calculation using mediapipe face mesh. (a) without glasses. (b) with glasses.

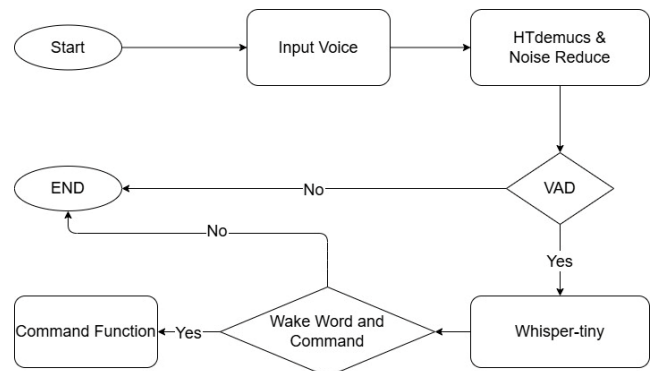


FIGURE 6. Flowchart of the voice interaction system using Whisper-tiny for VAD and wake word and command detection.

parameters for estimating fatigue and distraction tendencies [3], [12], [15], [16], [17]. When the fuzzy output exceeds a defined risk threshold, the system triggers an auditory alert via a buzzer. The decision-making process also considers outputs from the speech module to further validate the driver's verbal expressions, thereby enabling a consistent and robust multimodal driver monitoring process.

A. COMPARATIVE ANALYSIS OF EAR AND EAR-4 USING MACHINE LEARNING

To enhance the system's performance, we conducted a comparative analysis of the traditional six-point Eye Aspect Ratio (EAR) and a modified four-point Eye Aspect Ratio (EAR-4), both derived from eye landmarks extracted using MediaPipe. The evaluation was carried out on the NTHU Driver Drowsiness Detection Dataset, with classification accuracy assessed through two machine learning models: Random Forest and XGBoost. The conventional EAR relies on landmarks distributed across the upper and lower eyelids, together with horizontal corner points, to provide a holistic measure of eye openness. In contrast, the proposed EAR-4 approach streamlines the calculation by focusing solely on the deformation of the upper eyelid. This refinement not only reduces computational complexity but also improves sensitivity to fatigue-related eyelid dynamics, thereby offering a more efficient and targeted metric for drowsiness detection.

The Random Forest classifier is composed of multiple independent decision trees. While each tree shares the same structural framework of nodes, they are trained on different subsets of data, resulting in diverse outputs at the leaf level. Ultimately, the classifier aggregates the predictions of all trees, averaging their outcomes to produce a single, more robust and reliable result.

$$MSE = \frac{1}{N} \sum_{i=1}^N (f_i - y_i)^2 \quad (15)$$

where N is the total number of samples, f_i represents the predicted value of the model, y_i denotes the actual label.

$$Gini = 1 - \sum_{i=1}^N (p_i)^2 \quad (16)$$

$$Entropy = \sum_{i=1}^N -p_i \log_2 (p_i) \quad (17)$$

where p_i represents the proportion of samples belonging to class i at the given node, and N denotes the total number of categories.

The XGBoost belongs to the boosting family of algorithms. Unlike Random Forest, which builds trees independently, XGBoost constructs trees sequentially, with each new tree aiming to correct the residual errors of the previous ensemble. During optimization, XGBoost leverages both first-order and second-order derivative information, allowing for more precise gradient updates. In addition, it incorporates regularization terms to control model complexity, which helps reduce overfitting and enhances generalization performance.

$$L^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t(x_i)^2 \right] + \Omega(f_t) \quad (18)$$

where g_i represents the first-order gradient, h_i represents the second-order derivative.

In this study, image data from four representative subjects in the NTHU-DDD dataset were utilized, as illustrated through the video frame sequences in Fig. 7. The selected dataset comprises a total of 65,834 image frames, spanning diverse environmental conditions such as the presence or absence of prescription glasses. As summarized in Table 1, each frame is categorized as either a 'drowsy' (closed-eye) or 'non-drowsy' (open-eye) state. Out of the 65,834 total frames, 29,804 are labeled as open eyes, while 36,030 correspond to closed-eye states. To ensure a robust evaluation and prevent data leakage, the dataset was partitioned into a training set of 52,666 frames and a testing set of 13,168 frames, maintaining an approximate 80/20 distribution across all subjects. Subjects 2 and 3 contain a higher proportion of closed-eye frames, whereas Subject 4 shows the opposite trend. This initial split serves as a baseline evaluation, while subject-independent validation is discussed in Section V-E Future Work.

TABLE 1. Distribution of open-eye and closed-eye frames for each subject in the dataset.

Dataset Subject	Open Eyes (Frames)	Closed Eyes (Frames)	Train set (Frames)	Test Set (Frames)	Total Samples
1	9289	9584	15098	3775	15098
2	7834	10596	14744	3686	18430
3	8823	13087	17528	4382	21910
4	3858	2763	5296	1325	6621

We use this dataset to train and evaluate our models. From these data, we computed both the six-point EAR and the four-point EAR-4 features, as illustrated in Figs. 8-11. We examined the boxplots of the average EAR, as shown in Fig. 12. We observe that, compared with the conventional six-point EAR, EAR-4 shows a larger separation between the medians of the open-eye and closed-eye classes and a markedly smaller overlap in their interquartile ranges. This indicates that the four-point, upper-eyelid-focused design yields a cleaner decision boundary. Overall, EAR-4 is more sensitive to fatigue related to upper-eyelid dynamics and produces tighter within class distributions, which benefits subsequent classification.

To further validate the discriminative power of the features, a comparative evaluation of classification performance was conducted using two mainstream machine learning models: Random Forest (RF) and Extreme Gradient Boosting (XGB). These models were tasked with distinguishing between open and closed-eye states. The detailed quantitative results are compiled in Table 2, while trend comparisons are visualized in Fig. 12.

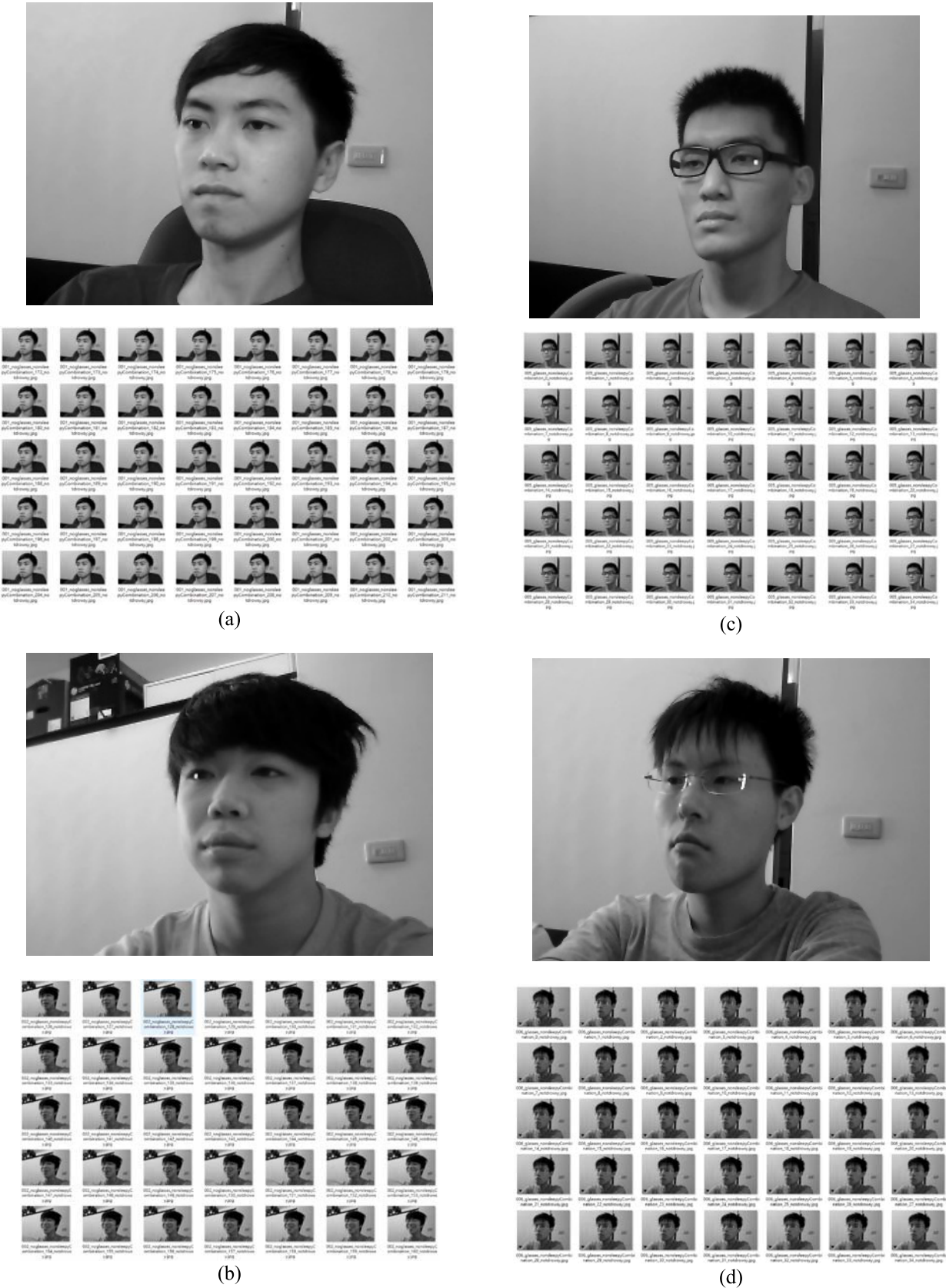


FIGURE 7. NTHU dataset.



FIGURE 8. Subject 1. Left: 6-point EAR. Right: 4-point EAR-4.



FIGURE 9. Subject 2. Left: 6-point EAR. Right: 4-point EAR-4.



FIGURE 10. Subject 3. Left: 6-point EAR. Right: 4-point EAR-4.



FIGURE 11. Subject 4. Left: 6-point EAR. Right: 4-point EAR-4.

The experimental data, as shown in Table 2, clearly indicates that for the majority of subjects, the EAR-4 feature achieves superior performance over EAR across key metrics, including accuracy, recall, and F1-score. This superiority is exemplified by the results for Subject 2; when the feature was switched from EAR to EAR-4 in the Random Forest model, the recall value saw a significant increase from 0.8560 to 0.8940, and the F1-score rose from 0.8130 to 0.8452. Importantly, this trend was consistently replicated with the XGB model. These findings provide compelling evidence that the proposed EAR-4 feature possesses enhanced discriminative potential, enabling more robust and precise differentiation between the two physiological states.

When comparing classifiers, XGB generally achieves higher recall, meaning it is more effective at reducing false negatives. This observation is also supported by the

confusion matrices in Fig. 13, where XGB produces fewer errors in the closed-eye class compared with RF. On the other hand, RF maintains a more balanced trade-off between precision and recall, thereby reducing the likelihood of false positives. Overall, the combination of EAR-4 with XGB delivers the best performance, achieving a balanced trade-off between accuracy and recall while maintaining the lowest error rate. These findings indicate that adopting the EAR-4 feature design can effectively enhance the robustness of driver drowsiness detection. In addition, Figs. 14–17 provide qualitative examples illustrating EAR-4 advantage over the conventional EAR: in identical frames, EAR-4 captures

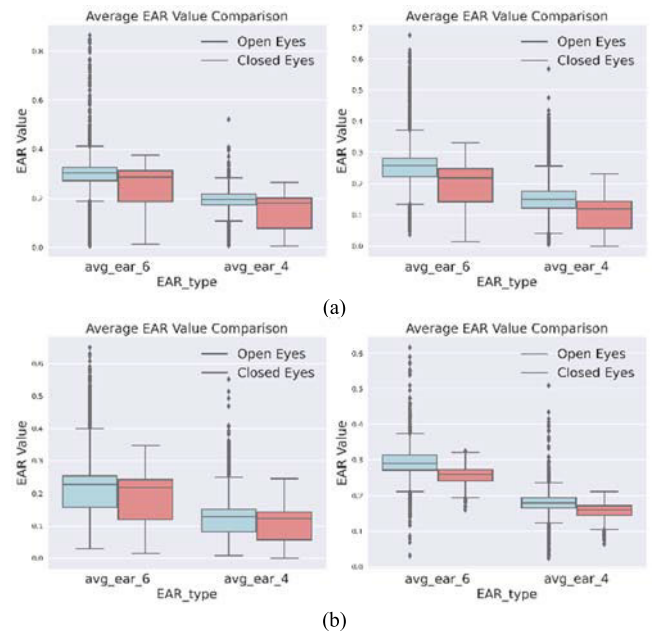


FIGURE 12. (a) Shows boxplots for Subjects 1–2, (b) shows those for subjects 3–4.

TABLE 2. Classification performance of ear and ear-4 mtehdos using random forest and XGBoost.

Subject	Model	Method	Accuracy	Recall	F1
1	RF	EAR	0.7313	0.7537	0.7402
1	RF	EAR-4	0.7446	0.7538	0.7498
1	XGB	EAR	0.7358	0.7636	0.7459
1	XGB	EAR-4	0.7610	0.7730	0.7666
2	RF	EAR	0.7737	0.8560	0.8130
2	RF	EAR-4	0.8117	0.894	0.8452
2	XGB	EAR	0.7848	0.8574	0.8208
2	XGB	EAR-4	0.8168	0.8862	0.8476
3	RF	EAR	0.7622	0.8792	0.8153
3	RF	EAR-4	0.7804	0.8765	0.8266
3	XGB	EAR	0.7692	0.8589	0.8164
3	XGB	EAR-4	0.7743	0.8674	0.8211
4	RF	EAR	0.7871	0.7902	0.7560
4	RF	EAR-4	0.8120	0.7848	0.7770
4	XGB	EAR	0.8075	0.7920	0.7745
4	XGB	EAR-4	0.8271	0.7848	0.7912

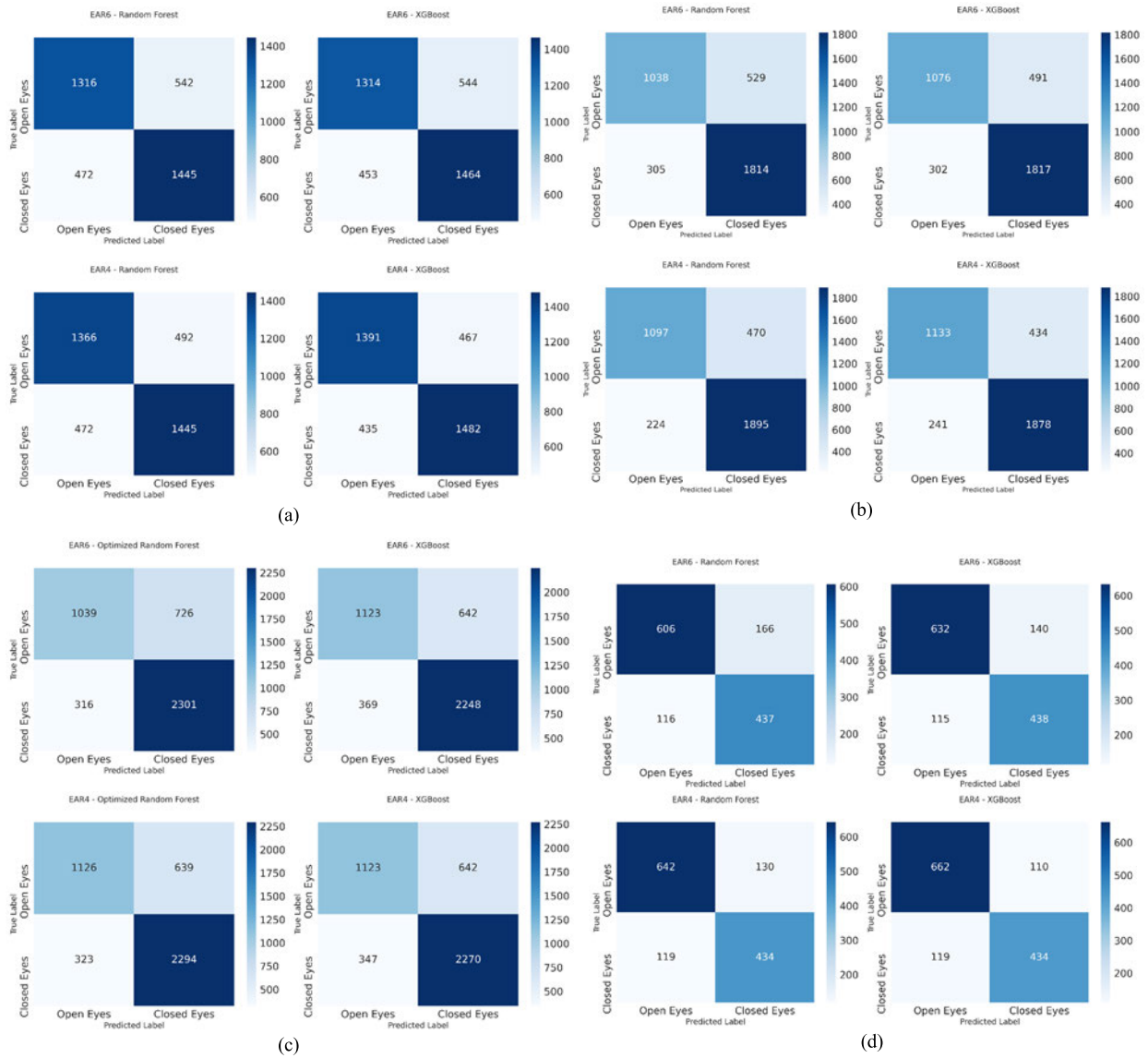


FIGURE 13. (a) Confusion matrix for Subject 1. (b) Confusion matrix for Subject 2. (c) Confusion matrix for Subject 3. (d) Confusion matrix for Subject 4.

subtle upper-eyelid changes more accurately and identifies partially closed eyes, whereas EAR tends to misclassify them as open.

B. HEAD POSE ESTIMATION TRACKING METHOD

To enhance the robustness of our driver monitoring system, particularly in scenarios where ocular features are unreliable (e.g., due to eye occlusion), we incorporated a Head Pose Estimation (HPE) module to capture non-visual attentional states. The module leverages MediaPipe FaceMesh to extract 2D facial landmarks from the video feed. Subsequently, a Perspective-n-Point (PnP) algorithm is employed to map these 2D points to a generic 3D facial model, thereby solving for the head's rotation and translation vectors relative to the camera, as detailed in equation (7). From the resulting rotation parameters, we derive the yaw and pitch angles,

which quantitatively represent the geometric orientation of the driver's head. These orientation cues are critical for identifying behaviors highly correlated with distraction and fatigue, such as head tilting or nodding. Based on empirical evaluation, we established calibrated thresholds to trigger a distraction alert. Specifically, an alert signal is sent to the vehicle's speaker if the yaw angle exceeds $\pm 13^\circ$ for more than 3 seconds (indicating a prolonged lateral glance) or if the pitch angle falls outside the range of -5° to $+10^\circ$ [5]. Yaw and pitch orientations align with the spatial specifications defined in [5] and [37]. The implementation and experimental outcomes of the HPE module are demonstrated in Fig. 18.

C. VOICE RECOGNITION METHOD

Following the architecture presented in Fig. 6, we developed a system for robust, real-time voice command recognition



FIGURE 14. Subject 1 EAR-4 marks the eye as partially closed, while EAR6 classifies it as open.



FIGURE 17. Subject 4 EAR-4 marks the eye as partially closed, while EAR6 classifies it as open.



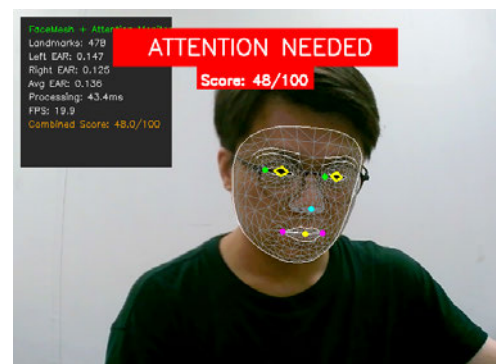
FIGURE 15. Subject 2 EAR-4 marks the eye as partially closed, while EAR6.



FIGURE 16. Subject 3 EAR-4 marks the eye as partially closed, while EAR6 classifies it as open.

within an in-vehicle environment. To address the adverse effects of music and background noise, the HTdemucs model is employed to perform source separation, which enhances

speech clarity and consequently improves the real-time performance of the Whisper-tiny ASR engine. The end-to-end framework strategically integrates this denoising component with voice activity detection (VAD), Whisper-tiny ASR, and wake-word verification. This hierarchical approach ensures that system functions are triggered exclusively by valid commands (e.g., “open the window,” “close the window”), effectively preventing false activations. The experimental outcomes of these commands are presented in Fig. 19.



(a)



(b)

FIGURE 18. HPE status detection. (a) The driver looks downward to the right. (b) The driver looks downward to the left.

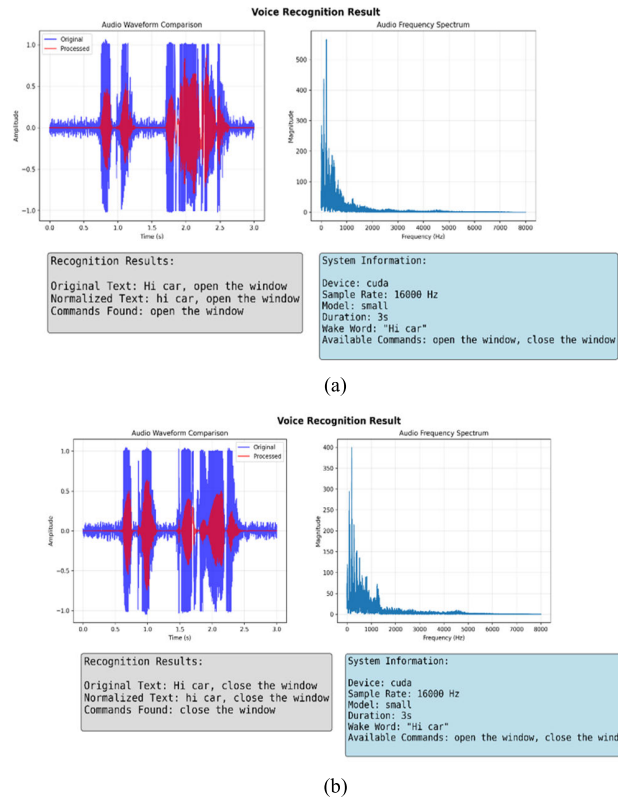


FIGURE 19. Voice recognition result. (a) open the window. (b) close the window.

D. BELIEVE RULE-BASED METHOD

The integration of multiple modalities within a belief rule-based framework markedly improves the robustness of fatigue detection. In particular, the system leverages three complementary inputs: (i) EAR4 eye features, which enable reliable discrimination between open and closed ocular states; (ii) head pose estimation (HPE), which quantifies deviations such as head tilting and downward posture; and (iii) Whisper-tiny speech recognition, which verifies the consistency between spoken intent and physiological indicators. By fusing these modalities, the framework consistently achieves superior performance over single-modality approaches across diverse driving scenarios.

As illustrated in Table 3, the first-stage rule base incorporates ocular features derived from the EAR4 metric together with head-pose parameters to map the driver’s condition into three drowsiness levels (High, Medium, Low), thereby generating an initial belief distribution. Building upon this, the second-stage rule base, as shown in Table 4, incorporates voice input as a confirmation signal and associates the inferred drowsiness states with beep-based alerts and concrete system actions. The primary contribution of this two-stage rule-based architecture lies in its ability to convert conventional passive fatigue detection into an active, integrated system, all while maintaining a crucial balance between interpretability and extensibility. This architectural approach effectively fuses visual information with voice

TABLE 3. Presents the first-stage belief rule base.

Rule	Driver State			Drowsiness State		
	Eye	Yaw	Pitch	High	Medium	Low
1	Open	Center	Center	0	0	1
2	Partial	Center	Center	0	0.7	0.3
3	Closed	Center	Center	1	0	0
4	Open	Left	Down	0	0.7	0.3
5	Open	Right	Down	0	0.7	0.3
6	Partial	Left	Down	0.7	0.3	0
7	Partial	Right	Down	0.7	0.3	0
8	Closed	Left	Down	1	0	0
9	Closed	Right	Down	1	0	0

TABLE 4. Illustrates the second-stage rule base.

Rule	Drowsiness State	Voice Input	Beep Alert	System Action
1	Low	NA	None	None
2	Medium/Low	I am ok	Single beep	Reduce Speed
3	High/Medium	No Worry	Double beep	Maintain Speed
4	High	Need help	Triple beep	Need assistance workflow

commands, not only enhancing the interpretability and consistency of driver state assessment but also exhibiting high portability and implementation flexibility. It is designed for deployment on edge device platforms to achieve real-time and efficient multi-modal decision-making capabilities.

E. EVALUATION OF SYSTEM PERFORMANCE

While existing Driver Monitoring Systems (DMS) vary in their methodologies and functionalities, the approach of using in-cabin cameras to monitor the driver’s head and eye states has been widely established as one of the most effective. The system proposed in this research falls into this category. However, a comprehensive evaluation of a DMS must account for multiple factors, and traditional passive monitoring often struggles to meet the demands of complex scenarios. To address this, our system introduces an active interaction capability to construct a more complete monitoring closed-loop, with its overall architecture depicted in Fig. 20.

The experimental results of this interactive mechanism are illustrated in Figs. 21-26. Initially, during the personalized calibration phase in Fig. 21, the system automatically acquires data over a 10 second period to establish a user-specific EAR baseline. Following calibration, the system enters a passive monitoring mode under normal driving conditions in Fig. 22(a), where it observes continuously without intervention. Fig. 22(b) illustrates the proposed Universal Fallback Mechanism, designed to mitigate initialization biases arising from sleep inertia or pre-existing fatigue. Acknowledging that the normal Eye Aspect Ratio (EAR) baseline for users with naturally smaller eye

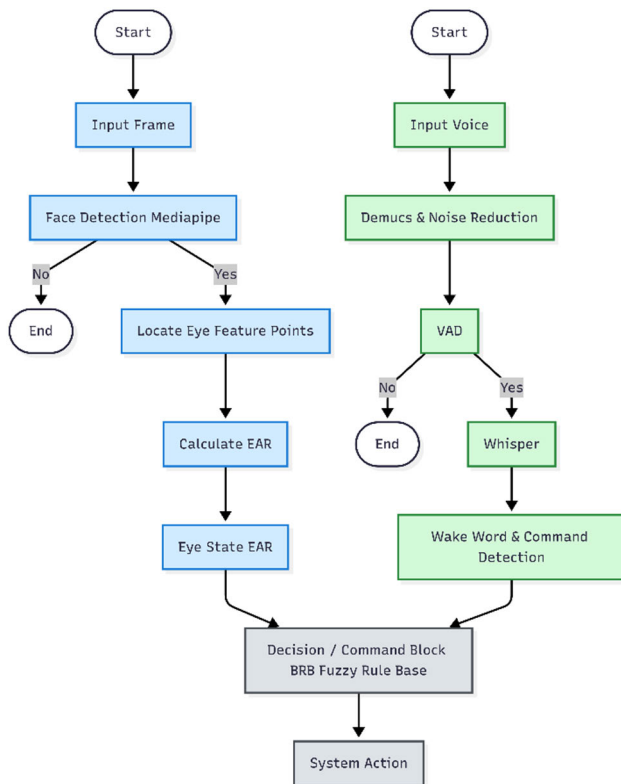


FIGURE 20. Architecture of the adaptive driver monitoring and interaction system.

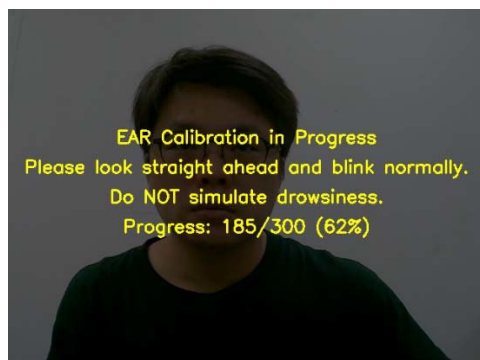
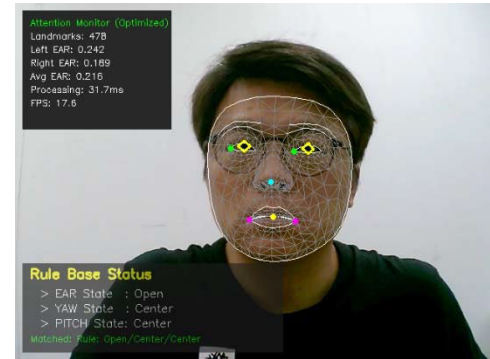


FIGURE 21. Personalized calibration phase.

characteristics typically ranges between 0.16 and 0.22, a distinction must be made between anatomical variations and drowsiness [6]. Consequently, if the mean EAR measured during the 10-second calibration phase falls below 0.16, the system classifies the data as unreliable due to potential drowsiness. In such scenarios, the personalized calibration result is rejected in favor of a redundancy protocol. This protocol enforces a default safety threshold of $EAR > 0.25$, aligning with the safety standards established in [5]. This mechanism significantly enhances the system's robustness against unreliable calibration data during the initialization phase. However, upon the detection of medium to high



(a)



(b)

FIGURE 22. System operational states: (a) Normal monitoring after successful calibration; (b) Universal fallback mechanism triggered by calibration failure.

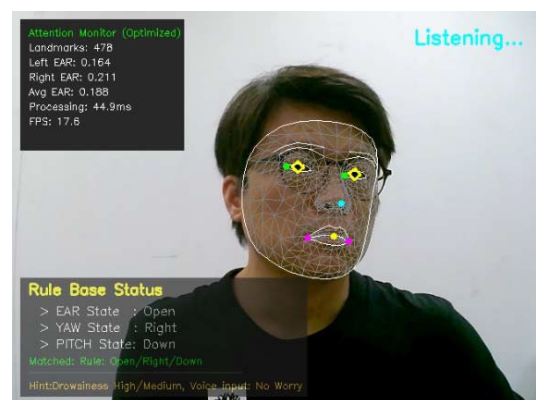


FIGURE 23. The system response in a High/Medium state based on the rule-based framework.

drowsiness indicators in Figs. 23-26, the system immediately transitions to an active mode, initiating voice monitoring for human-computer interactive verification. The system then awaits a verbal response from the user with keywords such as “no worry” or “need help” to complete the human-machine interactive verification. The system action layer functions as the execution stage of the multimodal framework, triggering in-vehicle responses based on the two-stage BRB rules in Table 4 and interfacing with external systems for additional safety measures. Tables 5 and 6 demonstrate that the proposed

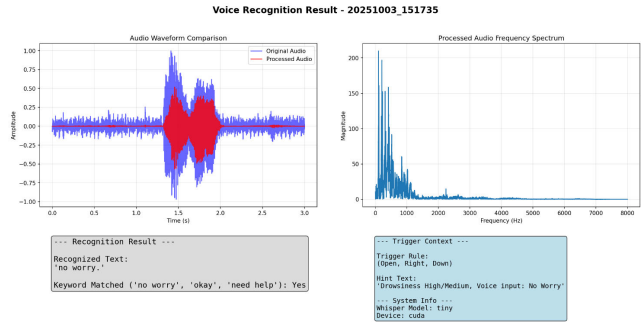


FIGURE 24. Successful recognition of the user’s “no worry” confirmation command.

system surpasses prior approaches while achieving real-time performance on edge platforms, underscoring its practical relevance [5].



FIGURE 25. The system response in a high state based on the rule-based framework.

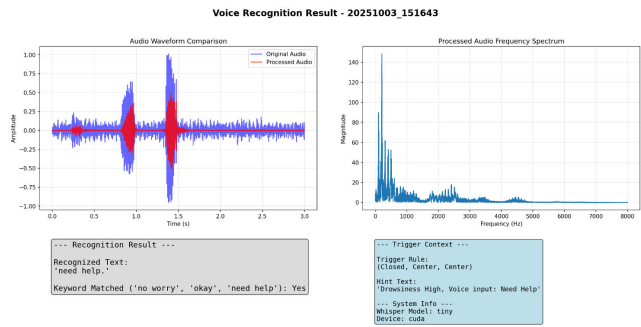


FIGURE 26. Successful recognition of the user’s “need help” confirmation command.

V. DISCUSSION

A. SAFETY CONSIDERATION OF ACTIVE VERBAL INTERACTION

This study has integrated these factors into the system design phase, particularly concerning driver responses during fatigue. From human factors engineering perspective, it is acknowledged that overly complex dialogue or high-

TABLE 5. Comparative analysis of different driver monitoring systems.

Research	Methodology	Detected Features	Detected Mechanism
Our work	CNN + BRB(Fuzzy Integration)	Drowsiness and Distraction	Multimodal Fusion(Facial + Speech)
[5]	CNN	Drowsiness and Distraction	Facial Features
[6]	CNN	Distraction	Facial Features
[7]	HOG + Linear SVM	Drowsiness	Facial Features
[18]	CNN + LSTM	Drowsiness and Distraction	Facial Spatiotemporal
[19]	CNN + RNN	Drowsiness	Facial Feature and Vehicle Dynamics
[20]	CNN	Drowsiness	Facial and Steering Wheel Features
[31]	DBN + Sliding Window	Drowsiness and Distraction	Facial Features

TABLE 6. Comparative analysis of different driver monitoring systems.

Research	Alert Mechanism	Frame Per Second	Device Implementation	Extended Functionality
Our work	Audio + Voice Prompt	25	YES	YES(Voice Command + LLM)
[5]	Audio + Seat Vibration	45	YES	YES
[6]	NO	34	YES	NO
[7]	NO	30	NO	NO
[18]	Audio	15	NO	NO
[19]	NO	15	NO	NO
[20]	NO	23	NO	NO
[31]	NO	30	NO	NO

level semantic reasoning could prove counterproductive by diverting the driver’s attention. However, evaluating this mechanism requires a clear distinction between high-burden social conversation and low-complexity, vigilance-maintaining interaction. The proposed system implements a highly simplified and structured lightweight protocol, necessitating only brief keywords rather than open-ended dialogue. Prior research indicates that for drivers entering a state of drowsiness, brief and low-difficulty vocal tasks such as simple word recognition or gamified challenges can effectively disrupt the monotony of long-duration driving and elevate neurophysiological arousal [32]. Since these interactions do not require deep semantic processing, cognitive load is maintained within a manageable threshold. To ensure minimal interference, an adaptive triggering mechanism is employed: voice intervention is initiated only when visual metrics, such as the Eye Aspect Ratio (EAR) or significant head pose deviations, fall below established safety thresholds. From a risk trade-off perspective, while proactive confirmation imposes a marginal increase in cognitive demand, such preventive intervention is logically justified and offers a significant

safety advantage over the unmitigated risks of drowsy driving.

B. GEOMETRIC SENSITIVITY AND TEMPORAL ROBUSTNESS OF EAR-4

As illustrated in Fig. 27, the EAR-4 metric exhibits an inherent geometric sensitivity to variations in head pitch. The observed reduction in EAR during downward head tilts is not merely a byproduct of perspective distortion; rather, it is intrinsically linked to physiological fatigue manifestations, such as nodding and involuntary ocular closure. Within the experimental sequence from approximately Frame 380 to 530, the EAR remains consistently below the 0.18 threshold, signifying a sustained eye-closed state. Subsequently, between Frames 650 and 790, the data captures a transition from prolonged closure to a downward head posture, reflecting a prototypical “nodding” drowsiness pattern. To mitigate potential misclassification arising from the volatility of instantaneous EAR values, the proposed system extracts high-order temporal features from the EAR time series. By incorporating descriptors such as blink duration and the proportion of sustained low-EAR segments, the model characterizes the longitudinal evolution of fatigue instead of relying on a static scalar threshold. This temporal analysis transforms EAR-4 from a simple geometric indicator into a robust representation of behavioral fatigue dynamics. Furthermore, to resolve the geometric ambiguity introduced by shifting head poses, a Head Pose Estimation (HPE) module is integrated as a stabilizing complementary cue. By explicitly modeling head orientation in conjunction with ocular features, this multi-cue architecture ensures a resilient assessment of driver state. Consequently, the system maintains high diagnostic reliability even when facial landmarks undergo significant perspective distortion due to extreme head movements.

C. TRADE-OFF BETWEEN MODEL AND REAL-TIME ROBUSTNESS

As illustrated in Fig. 28, the real-time capability of the proposed framework on the Jetson Orin Nano (8GB, 40 TOPS)



FIGURE 28. Jetson Orin Nano Developer Kit utilized as the primary edge computing platform [35].

is sustained through a prioritized task-scheduling strategy [33], [35]. By leveraging an asynchronous event-driven architecture, the system prioritizes the continuous execution of the visual monitoring backbone while offloading intensive audio processing tasks to background threads. This design prevents computational bottlenecks and ensures that high-frequency EAR and head-pose tracking remain uninterrupted, thereby optimizing resource allocation for maximum operational efficiency. To ensure seamless real-time operation, Fig. 29 details the system’s asynchronous execution flow and its task-decoupling mechanism. Our architectural design prioritizes the main execution thread for continuous visual monitoring, where high-frequency tasks such as MediaPipe facial landmark extraction and the geometric computation of EAR and head pose are processed without interruption. In contrast, the more computationally intensive speech verification pipeline is treated as an event-triggered process. By offloading these audio tasks to a low-priority background thread only when speech activity is detected, we effectively shield the critical visual inference loop from potential processing spikes. This asynchronous strategy provides robust latency isolation, ensuring that the system maintains its target 25 FPS even during active voice interaction. To provide a quantitative basis for our architectural design, Fig. 30 summarizes the empirical profiling results of the integrated audio processing modules. As illustrated in Fig. 30(a), the HTdemucs model asked with real-time speech enhancement operates at a normalized computational complexity of 15.24 GFLOPs while maintaining a compact footprint of 41.98 million parameters [34]. Complementing this, the Whisper-tiny ASR module (shown in Fig. 30(b)) exhibits an exceptionally low resource demand, requiring a mere 0.17 GFLOPs across its 37.18 million parameters [23]. The modest cumulative load of these models validates our strategy of utilizing lightweight yet robust architectures, ensuring the system remains responsive on edge hardware without exhausting the platform’s computational budget. It should be noted that Whisper-tiny is employed in this framework for short wake words and predefined command phrases rather than full-scale speech transcription. Its reported WER of 28.8% on the Common Voice 9 dataset [23] reflects a deliberate design trade-off favoring computational efficiency and low-latency interaction.



FIGURE 27. Temporal variations of EAR and head pitch under simulated drowsiness.

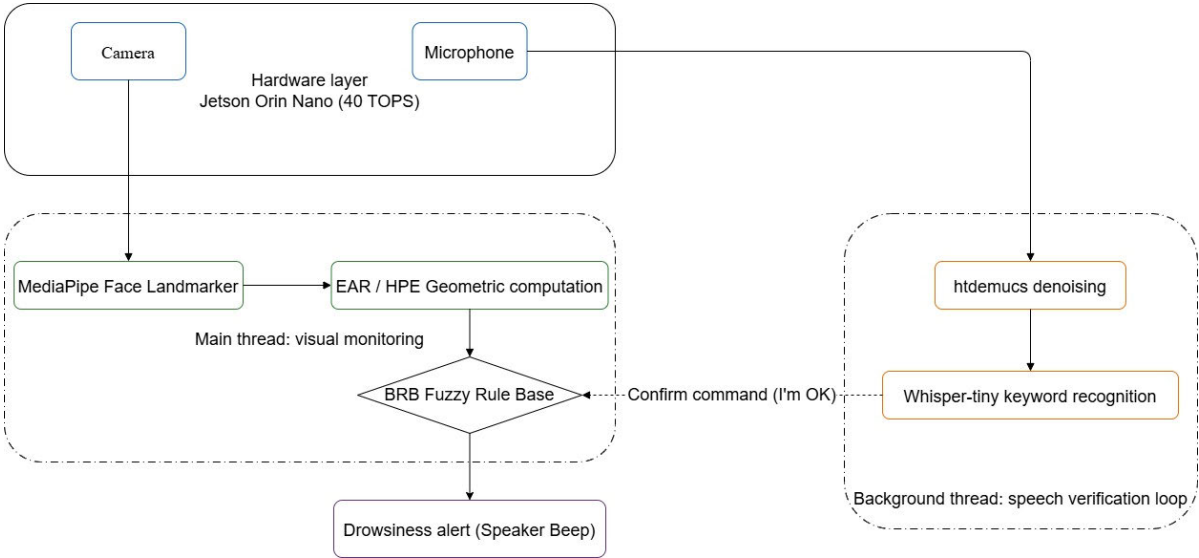
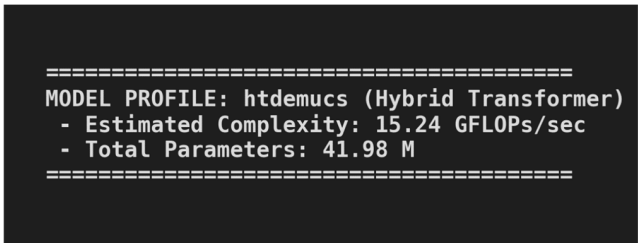
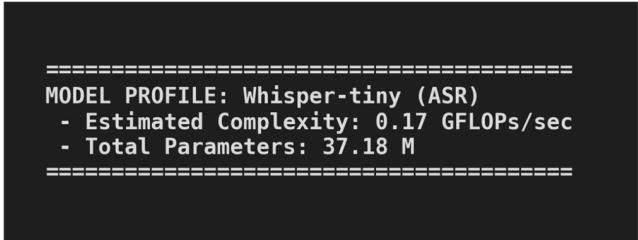


FIGURE 29. System execution flow and task decoupling strategy on the Jetson Orin Nano platform.



(a)



(b)

FIGURE 30. Computational profiling of the audio processing modules. (a) HTdemucs speech enhancement model. (b) Whisper-tiny ASR model.

D. CLASSIFICATION ROBUSTNESS AND FEATURE EFFICIENCY ANALYSIS

To further validate the practical utility of the simplified EAR-4 feature, its discriminative performance was rigorously benchmarked against the standard EAR using a machine learning-based evaluation pipeline, as illustrated in Fig. 31. As detailed in the systematic workflow, the study applied Random Forest (RF) and XGBoost classifiers to categorize eye states (open vs. closed) across datasets of varying scales. The integration of these classifiers serves a dual purpose: Random Forest provides a robust baseline for evaluating feature stability under noise, while XGBoost leverages gradient boosting to capture non-linear relationships within the

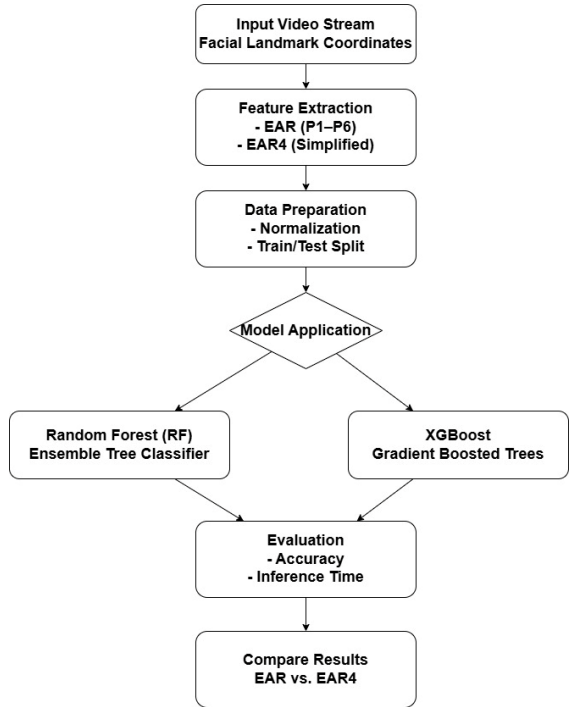


FIGURE 31. Systematic evaluation workflow for eye state classification using random forest (RF) and XGBoost algorithms.

temporal EAR data with high precision. As detailed in Section IV-A, EAR-4 reduces FPR compared to standard EAR by isolating upper-eyelid deformation. This enables precise separation of natural blinks from early drowsiness, ensuring high reliability with minimal driver distraction.

E. FUTURE WORK

While this study primarily focuses on eye-closure-centric fatigue modeling and real-time system validation, future work

will further extend the experimental scope to enhance generalization across datasets. The NTHU-DDD dataset is particularly suitable for this study due to its rich annotations of eye-closure behaviors under controlled yet diverse driving scenarios, which are well aligned with the proposed EAR-4 design and temporal fatigue analysis. Eye-closure focused datasets have been commonly adopted in prior driver drowsiness studies to validate visual fatigue indicators under controlled experimental settings [36].

VI. CONCLUSION

This research has proposed an active Adaptive Driver Monitoring and Assistance System (ADMAS) that effectively enhances driver alertness through real-time interaction, a significant advancement over traditional passive monitoring. The system's core is a multi-modal decision-making mechanism utilizing a Fuzzy Belief Rule Base to integrate two key information streams: a visual analysis stream employing an innovative EAR-4 feature, which achieves a high recall of 89.4% in classification tasks; and an efficient voice processing stream that employs HTdemucs and Whisper-tiny models for precise command recognition and active communication. The entire system has been successfully deployed on an NVIDIA Orin Nano edge device, operating stably in real-time at 25 FPS, thus validating its high feasibility and practical value for real-world applications.

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